Topic: Mixing problems

Question: A tank contains $1{,}000$ L of water and 10 kg of dissolved salt. Fresh water is filling the tank at 10 L/min, while the tank drains at 5 L/min. Assuming the solution stays perfectly mixed, how much salt is in the tank after t minutes?

Answer choices:

A $y = 5e^{-\frac{1}{100}t}$

B $y = 5e^{\frac{1}{100}t}$

C $y = \dfrac{2{,}000}{200 + t}$

D $y = \dfrac{10}{200 + t}$

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Solution: C

The concentration of salt in the freshwater entering the tank is $C_1 = 0$ kg/min, while the fill rate is $r_1 = 10$ L/min.

With a fill rate of 10 L/min and a drain rate of 5 L/min, the initial $1{,}000$ L in the tank is gaining 5 L every minute, so the concentration of salt draining from the tank is $C_2 = y/(1{,}000 + 5t)$ kg/L, while the drain rate is $r_2 = 5$ L/min.

Plugging these values into the differential equation gives

$$\frac{dy}{dt} = C_1 r_1 - C_2 r_2$$

$$\frac{dy}{dt} = (0)(10) - \left(\frac{y}{1{,}000 + 5t}\right)(5)$$

$$\frac{dy}{dt} = -\frac{5y}{1{,}000 + 5t}$$

$$\frac{dy}{dt} = -\frac{y}{200 + t}$$

This is now a separable differential equation, so we'll separate variables and integrate both sides.

$$\int \frac{1}{y}\, dy = \int -\frac{1}{200 + t}\, dt$$

$$\ln|y| = -\ln|200 + t| + C$$

Solve for y.

$$e^{\ln|y|} = e^{-\ln|200+t|+C}$$

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$$|y| = Ce^{-\ln|200+t|}$$

$$y = Ce^{-\ln|200+t|}$$

Because we're talking about a real-world problem in which time t will never be negative, $200 + t$ will always be positive, so we can take away the absolute value.

$$y = Ce^{-\ln(200+t)}$$

By laws of logarithms, the negative sign in front of the logarithm becomes the exponent on the argument.

$$y = Ce^{\ln(200+t)^{-1}}$$

$$y = C(200 + t)^{-1}$$

$$y = \frac{C}{200 + t}$$

The tank initially contained 10 kg of dissolved salt, so we'll plug $y(0) = 10$ into this general solution.

$$10 = \frac{C}{200 + 0}$$

$$C = 2{,}000$$

So the amount of salt in the tank at any time t can be modeled by

$$y = \frac{2{,}000}{200 + t}$$

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Topic: Mixing problems

Question: A tank contains $1{,}000$ L of water and 15 kg of dissolved salt. Fresh water is filling the tank at 10 L/min, while the tank drains at 20 L/min. Assuming the solution stays perfectly mixed, how much salt is in the tank after 10 minutes?

Answer choices:

A 12.40 kg

B 12.15 kg

C 16.75 kg

D 13.60 kg

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Solution: B

The concentration of salt in the fresh water entering the tank is $C_1 = 0$ kg/min, while the fill rate is $r_1 = 10$ L/min.

With a fill rate of 10 L/min and a drain rate of 20 L/min, the initial $1{,}000$ L of water in the tank is decreasing by 10 L every minute, so the concentration of salt draining from the tank is $C_2 = y/(1{,}000 - 10t)$ kg/L, while the drain rate is $r_2 = 20$ L/min.

Plugging these values into the differential equation gives

$$\frac{dy}{dt} = C_1 r_1 - C_2 r_2$$

$$\frac{dy}{dt} = (0)(10) - \left(\frac{y}{1{,}000 - 10t}\right)(20)$$

$$\frac{dy}{dt} = -\frac{20y}{1{,}000 - 10t}$$

$$\frac{dy}{dt} = -\frac{2y}{100 - t}$$

This is now a separable differential equation, so we'll separate variables and integrate both sides.

$$\int \frac{1}{y}\, dy = \int -\frac{2}{100 - t}\, dt$$

$$\ln|y| = 2\ln|100 - t| + C$$

Solve for y.

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$$e^{\ln|y|} = e^{2\ln|100-t|+C}$$

$$|y| = Ce^{2\ln|100-t|}$$

$$y = Ce^{2\ln|100-t|}$$

By laws of logarithms,

$$y = Ce^{\ln|100-t|^2}$$

$$y = C|100-t|^2$$

$$y = C(100-t)^2$$

The tank initially contained 15 kg of dissolved salt, so we'll plug $y(0) = 15$ into this general solution.

$$15 = C(100-0)^2$$

$$15 = 10{,}000C$$

$$C = \frac{3}{2{,}000}$$

So the amount of salt in the tank at any time t can be modeled by

$$y = \frac{3}{2{,}000}(100-t)^2$$

To find the amount of salt in the tank after 10 minutes, we'll evaluate at $t = 10$.

$$y(10) = \frac{3}{2{,}000}(100-10)^2$$

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$$y(10) = \frac{3(8{,}100)}{2{,}000}$$

$$y(10) = \frac{243}{20}$$

$$y(10) = 12.15 \text{ kg}$$

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Topic: Mixing problems

Question: A tank contains $2{,}000$ L of water and 25 kg of dissolved salt. Fresh water is filling the tank at 10 L/min, while the tank drains at 10 L/min. Assuming the solution stays perfectly mixed, how much salt is in the tank after 5 minutes?

Answer choices:

A 25.6 kg

B 25.1 kg

C 24.9 kg

D 24.4 kg

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Solution: D

The concentration of salt in the fresh water entering the tank is $C_1 = 0$ kg/min, while the fill rate is $r_1 = 10$ L/min.

With a fill rate of 10 L/min and a drain rate 10 L/min, the total volume in the tank stays, which means the volume at any time t is $2{,}000$ L.

Plugging these values into the differential equation gives

$$\frac{dy}{dt} = C_1 r_1 - C_2 r_2$$

$$\frac{dy}{dt} = (0)(10) - \left(\frac{y}{2{,}000}\right)(10)$$

$$\frac{dy}{dt} = -\frac{10y}{2{,}000}$$

$$\frac{dy}{dt} = -\frac{y}{200}$$

This is now a separable differential equation, so we'll separate variables and integrate both sides.

$$\int \frac{1}{y}\, dy = \int -\frac{1}{200}\, dt$$

$$\ln|y| = -\frac{1}{200}t + C$$

Solve for y.

$$e^{\ln|y|} = e^{-\frac{1}{200}t + C}$$

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$$|y| = Ce^{-\frac{1}{200}t}$$

$$y = Ce^{-\frac{1}{200}t}$$

The tank initially contained 25 kg of dissolved salt, so we'll plug $y(0) = 25$ into this general solution.

$$25 = Ce^{-\frac{1}{200}(0)}$$

$$C = 25$$

So the amount of salt in the tank at any time t can be modeled by

$$y = 25e^{-\frac{1}{200}t}$$

To find the amount of salt in the tank after 5 minutes, we'll evaluate at $t = 5$.

$$y = 25e^{-\frac{1}{200}(5)}$$

$$y = 25e^{-\frac{1}{40}}$$

$$y \approx 24.4 \text{ kg}$$

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